A comparative review of generalizations of the extreme value distribution

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Abstract

The extreme value distribution, also known as the Gumbel distribution, is widely applied for extreme value analysis but has certain drawbacks in practice because it is a non heavy-tailed distribution and is characterized by constant skewness and kurtosis. Our goal is to present a literature review of the distributions that contain the extreme value distribution embedded in them and to identify those that have flexible skewness and kurtosis and those that are heavy-tailed. The generalizations of the extreme value distribution are described and compared using an application to a wind speed data set and Monte Carlo simulations. We show that some distributions suffer from overparameterization and coincide with other generalized Gumbel distributions with a smaller number of parameters, i.e., are non-identifiable. Our study suggests that the generalized extreme value distribution and a mixture of two extreme value distributions should be considered in practical applications.

Keywords: Generalized extreme value distribution; Gumbel distribution; Heavy-tailed distribution; Mixture of extreme value distributions.

1. Introduction

Extreme value data usually exhibit excess kurtosis and/or heavy right tails. This is particularly common in environmental data, e.g., maximum water level, maximum wind speed, spatial and temporal variability of turbulence, daily maximum ozone measurement, and largest lichen measurements. The Gumbel distribution is frequently used to model extreme values. However, its skewness and kurtosis coefficients are constant, and its right tail is light. Generalizations of the Gumbel distribution with flexible skewness and kurtosis coefficients could provide a better fit for extreme value data. We present a comparative review of distributions that contain the Gumbel distribution as a special or limiting case. Since we may draw different conclusions from nonidentifiable models, we distinguish between the identifiable and nonidentifiable models. Some distributions suffer from overparameterization and coincide with other generalized Gumbel distributions with smaller number of parameters. We limit our study to the identifiable family of distributions only.

We compare their coefficients of skewness and kurtosis (which are invariant under location-scale transformations) that are primarily controlled by the extra parameters. We highlight those that can achieve high values of skewness and kurtosis with a heavy right tail.

To study the tail behavior of the distributions, we also employ the regular variation theory (de Haan, 1970) and a criterion proposed by Rigby et al. (2014).

Additionally, we conduct a comprehensive simulation study to evaluate the flexibility of each selected distribution in fitting data sets generated from the Gumbel distribution and from its different generalizations. We compare the different distributions through the analysis of a data set on the maximum monthly wind speed in West Palm Beach, Florida, for years 1984-2014.

2. The Gumbel distribution and its generalizations

The Gumbel distribution is one of the three possible limiting laws of the standardized maximum of independent and identically distributed random variables (Gnedenko, 1943). This distribution is also known as the extreme value (EV(μ, σ)) or type I extreme value distribution.

Table 1 gives distributions that contain the Gumbel distribution as a special or limiting case; the nonidentifiable distributions are marked with an asterisk (*). The coefficients of skewness and kurtosis of the Gumbel distribution are constant ($\gamma_{1;\mathrm{EV}} = 1.14$ and $\gamma_{2;\mathrm{EV}} = 5.4$, respectively), i.e., parameter independent. The other distributions have a range of values of skewness and kurtosis coefficients which are primarily controlled by the extra parameters. The GEV distribution is the only one that allows unlimited skewness and kurtosis. From the regular variation theory (de Haan, 1970), the generalized extreme value distribution GEV(μ,σ,α) has a heavy right tail with tail index α when $\alpha > 0$ (Fréchet family). It has a non-heavy right tail when $\alpha = 0$ (Gumbel family). The other identifiable distributions addressed in this paper are all non-heavy right tailed distributions. Hence, among the identifiable distributions addressed in this work, the GEV distribution is the only one with a heavier right tail with respect to the Gumbel distribution under the tail index approach. To distinguish among the generalized distributions we used the Rigby et al. (2014) criterion. The criterion splits the distributions in three types, I, II and III, in decreasing order of heaviness of the tails. The right tail of the GEV distribution with $\alpha > 0$ is of type I. If $\alpha > 1$, the GEV distribution has a heavier right tail than the Cauchy distribution. If $\alpha > 1/2$, the right tail heaviness of the GEV distribution is greater than that of the Student-t distribution with two degrees of freedom, which is uncommon in real data. The Gumbel distribution and all of the other generalizations are of type II. The EGu, EGa, and GLIV distributions have heavier right tail than the Gumbel distribution when $\alpha < 1$. The TEV, GGu3 and TCEV distributions have lighter right tail than the GEV, EGu, EGa, and GLIV distributions. The TEV distribution when $\alpha < 0$ and the TCEV$(0, 1, 10, 5, \alpha)$ distribution have heavier right tail than the Gumbel distribution. The right tail of the GGu3 distribution is lighter than that of the Gumbel distribution.

3. Monte Carlo simulation results

We next compare the ability of the Gumbel distribution and its generalizations to model data taken from different distributions. To this end, we present a Monte Carlo simulation study in which 10,000 samples of size 500 were generated from and modeled with each of the identifiable distributions considered in this paper. For generating the data, we set $\mu = 0$ and $\sigma = 1$ (for the TCEV distribution $\mu_1 = 10$ and $\sigma_1 = 5$). The remaining parameters were chosen in such a way that the .99 quantile is near eight whenever possible. Table 2 presents distributions from which the samples were drawn and their .99 quantile and kurtosis.
As measures of model adequacy, we use the Akaike information criterion (AIC) and two modified Anderson Darling statistics (ADR and AD2R). AIC is a measure of dissimilarity between two distributions over the support; smaller AIC indicates a better fit. ADR and AD2R (Luceño, 2005, Table 2 and B.1) are sensitive to the lack of fit in the right tail of the distribution. AD2R puts more weight in the right tail than ADR. Smaller values of ADR and AD2R are indicative of a better fit.
A characteristic that is often of interest in extreme data modeling is the return level. The return level with return period $1/p$ is the quantile x_{1-p}, and it is interpreted as the value that we expect to be exceeded once every $1/p$ periods on average. To evaluate the quantile goodness of fit, we compute the .99 quantile discrepancy, which is defined as the difference between the .99 quantile of the fitted model and the .99 quantile of the distribution from which samples are generated divided by the latter.
The maximum likelihood estimates of the parameters were obtained by numerically maximizing the log-likelihood function. For the maximization procedure, we used the nonlinear quasi-Newton BFGS method

Table 1: Generalizations of the Gumbel distribution.

Distribution	Proposed by
Generalized extreme value GEV(μ,σ,α)	Jenkinson (1955)
Type IV generalized logistic GLIV(μ,σ,α)	Prentice (1975)
Two-component extreme value TCEV$(\mu,\sigma,\mu_1,\sigma_1,\alpha)$	Rossi et al. (1984)
Three parameter exponential-gamma EGa(μ,σ,α)	Ojo (2001)
Exponentiated Gumbel EGu(μ,σ,α)	Nadarajah (2006)
Transmuted extreme value TEV(μ,σ,α)	Aryal & Tsokos (2009)
Generalized three-parameter Gumbel GGu3(μ,σ,α)	–
Generalized type I extreme value GGu$(\mu,\sigma,\alpha,\beta)$ *	Dubey (1969)
Exponential-gamma ExpGamma$(\mu,\sigma,\alpha,\beta)$ *	Adeyemi & Ojo (2003)
Beta Gumbel BG$(\mu,\sigma,\alpha,\beta)$ *	Nadarajah & Kotz (2004)
Kummer beta generalized Gumbel KBGGu$(\mu,\sigma,\alpha,\beta,\gamma)$ *	Pescim et al. (2012)
Kumaraswamy Gumbel KumGum$(\mu,\sigma,\alpha,\beta)$ *	Cordeiro et al. (2012)
Exponentiated generalized Gumbel EGGu$(\mu,\sigma,\alpha,\beta)$ *	Cordeiro et al. (2013)

Table 2: Distributions, .99 quantile and kurtosis

	EV	GEV	EGu	TEV	EGa	GLIV	TCEV
Parameters	–	$\alpha = 0.22$	$\alpha = 0.6$	$\alpha = -0.99$	$\alpha = 0.6$	$\alpha = 0.55$	$\alpha = 0.0125$
						$\beta = 10$	$\mu_1 = 10$, $\sigma_1 = 5$
$x_{.99}$	4.60	7.96	7.68	5.29	7.85	7.97	8.03
Kurtosis	5.40	90.16	6.66	5.39	6.59	6.60	5.26

with numerical derivatives (see, e.g., Press et al., 1992), and a method that implements a sequential quadratic programming technique to maximize a non-linear function subject to non-linear constraints, similar to Algorithm 18.7 in Wright & Nocedal (1999). These methods are individually implemented in the functions MaxBFGS and MaxSQP in the matrix programming language `Ox` (Doornik, 2013). For the GEV distribution we employed two estimation methods, namely maximum likelihood (GEV-MLE) and probability weighted moments (GEV-PWM) (Castillo et al., 2005, Section 5.3). For the TEV distribution, we used the profile log-likelihood function for the parameter α.

Figures 1 and 2 present the boxplots of AIC, ADR, AD2R, and the quantile discrepancies of the fitted models. For these figures, the samples were generated from a standard Gumbel distribution, i.e., EV$(0, 1)$, and a generalized extreme value GEV$(0, 1, 0.22)$ distribution, respectively. The figures corresponding to the cases where the samples were generated from the exponentiated Gumbel EGu$(0, 1, 0.6)$, transmuted extreme value TEV$(0, 1, -0.99)$, three-parameter exponential-gamma EGa$(0, 1, 0.6)$, type IV generalized logistic GLIV$(0, 1, 0.55, 10)$ and two-component extreme value TCEV$(0, 1, 10, 5, 0.0125)$ distributions were also obtained but are not shown for saving space.

For the Gumbel distribution samples (Figure 1), the boxplots of AIC are quite similar and suggest that all generalizations of the Gumbel distribution can suitably fit Gumbel distributed data. The goodness of fit at the right tail, illustrated by the boxplots of ADR, are also quite similar except for the Gumbel and the GLIV distributions. For the Gumbel distribution, the median and the dispersion are bigger than for the other distributions. The GLIV fit is poor in the right tail, which is consistent with its .99 quantile plot , that suggests that a small difference in the estimated parameter α produces a significant difference in the upper quantiles. This characteristic makes the right tail goodness of fit to be dependent on the precision adopted for the parameter estimation. Boxplots of AD2R emphasize right-tail lack of fit. The boxplots of AD2R in Figure 1 are similar except for the GLIV fit, whose interquartile range (IQR) is the largest. The GEV-MLE right tail fit seems to be slightly better than the others. The boxplots of 0.99 quantile discrepancy show that the median is close to zero for all the distributions. The EV fit exhibits the smallest amplitude.

For the GEV distribution samples (Figure 2), the boxplots of AIC of the GEV-MLE and GEV-PWM fits present the smallest medians and the GEV-PWM fit exhibits the smallest amplitude. The boxplots of ADR and AD2R highlight the GEV-PWM fit as the best right-tail fit. The boxplots of quantile discrepancy show that the GEV-PWM, GEV-MLE and TCEV fits have medians closest to zero, and all the others underestimate the .99 quantile, markedly the EV and TEV fits.

Hereafter we analyze the fits when the data were generated from the other distributions. The boxplots of AIC are similar to those in Figure 1, except when the data were generated from the TCEV distribution. In this case, the TCEV distribution presents the smallest median value. The boxplots of ADR for the TCEV fit exhibit the smallest medians and IQR followed by the GEV-PWM fit.

The boxplots of AD2R reveal that the GEV-MLE and GEV-PWE fits are among the best fits regardless of from which distribution the data were generated. When generating the data from the TCEV distribution, all the distributions underestimate the .99 quantile and the median of the GEV-PWM fit is the closest to zero. Summing up, the GEV-PWM, TCEV, and GEV-MLE fits are among the best fits in all of the simulated settings. The simulation results suggest that the GEV and TCEV distributions provide the best right-tail fits.

4. Application to a wind speed data set

We analyze data on the maximum monthly peak gust wind speed (mph) in West Palm Beach, Florida (USA) for years 1984-2014, with $n = 371$ observations. We fitted the different identifiable models described above to the data.

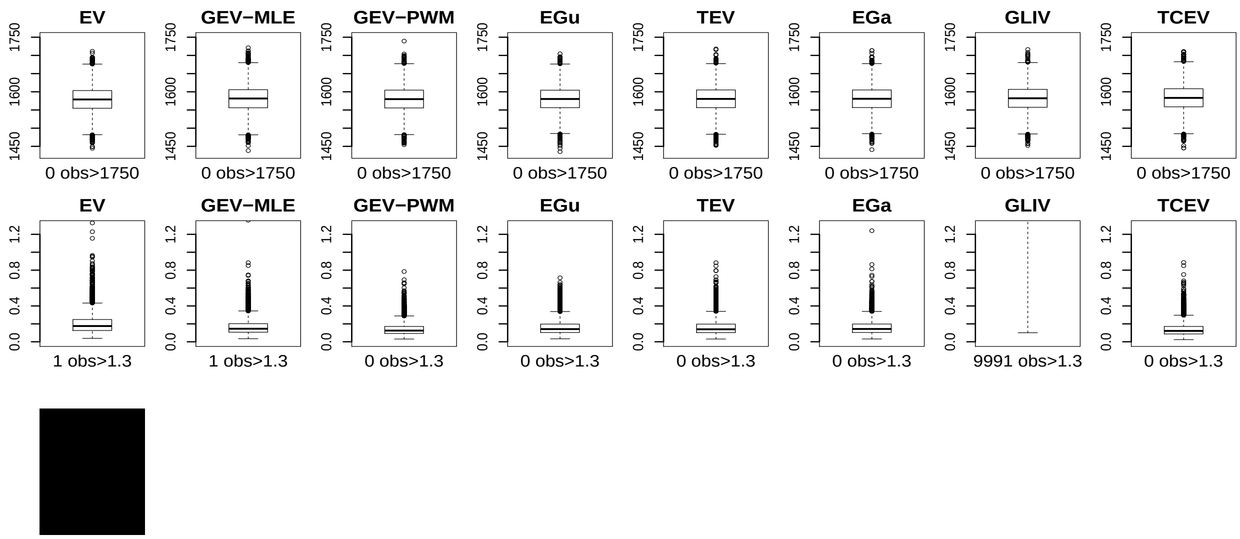

Figure 1: Boxplots of AIC (first row), ADR (second row), AD2R (third row) and quantile discrepancies (fourth row) - random samples generated from EV.

A summary of the goodness of fit measures is given in Table 3. Unlike the other distributions, the GGu3 distribution does not exhibit better fit relative to the Gumbel distribution according to the AIC criterion. The GEV-MLE fit produces the smallest AIC (2522.80), followed by the GEV-PWM (2523.03), while the GGu3 distribution exhibits the largest AIC. The lowest ADR value is achieved by the TCEV distribution (0.34), followed by the GEV-PWM fit (0.41) and GEV-MLE fit (0.42). The Gumbel and GGu3 distributions produce the greatest ADR (0.56). The AD2R measure highlights the Gumbel distribution (117.35) and the GGu3 distribution (117.38) as the worst fits. It points to the TCEV distribution (2.84) and the GEV-PWM fit (4.70) as the best fits. All the goodness of fit measures reveal that the Gumbel distribution is not the best choice for this data set. Taking all of the goodness of fit criteria into account, we conclude that the GEV and TCEV distributions produce the best fits.

Table 3: Goodness of fit measures - wind speed data.

	EV	GEV MLE	GEV PWM	EGu	TEV	GGu3	EGa	GLIV	TCEV
$\ell(\widehat{\theta})$	-1261.02	-1258.40	-1258.51	-1259.47	-1259.43	-1261.02	-1259.40	-1259.21	-1257.65
AIC	2526.04	2522.80	2523.03	2524.94	2524.86	2528.04	2524.79	2526.43	2525.30
ADR	0.56	0.42	0.41	0.49	0.43	0.56	0.48	0.44	0.34
AD2R	117.35	6.54	4.70	31.69	42.02	117.38	30.01	25.07	2.84

Figure 3 displays the profile log-likelihood function for the additional parameter(s) of the fitted models. Note the absence of concavity in the profile log-likelihood function for the EGu and EGa models, which casts doubt on the approximate standard error for the MLE of α. The profile log-likelihood function for the TEV model exhibits an inflection point, a local minimum, and two local maxima, and hence maximization can converge to a local maximum depending on the initial value. The profile log-likelihood function for the GGu3 model is increasing and flat for large values of α. Hence, the estimate of α depends on the numerical

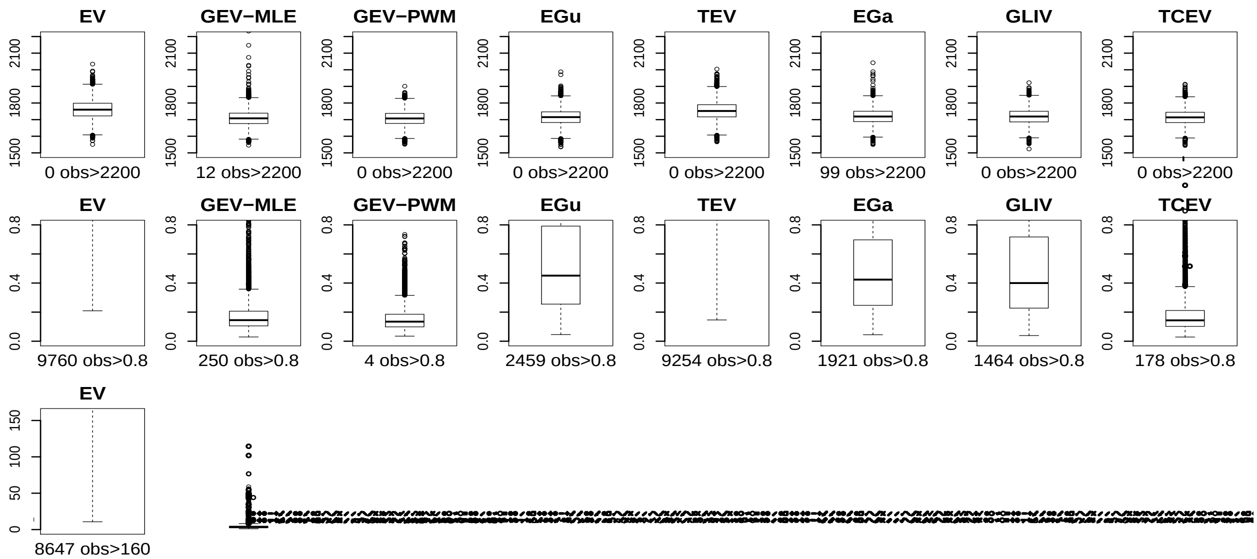

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Figure 2: Boxplots of AIC (first row), ADR (second row), AD2R (third row) and .99 quantile discrepancies (fourth row) - Random samples generated from GEV.

precision specified for the maximization algorithm, and the standard error estimate is very large. The profile log-likelihood function for the GLIV model also grows slowly in the β parameter direction, but its β estimate is not as large as for the GGu3 model. Therefore, there is no difficulty with standard error estimation despite its large size due to lack of concavity of the curve. The profile log-likelihood function for the TCEV model presents an inflection point, likely due to numerical issues, which appears not to disturb the parameter estimation; however, it is a concave function. The profile log-likelihood function for the GEV model is well behaved with no inflection point, multimodality or lack of concavity.
The boxplots in the Figure 4 are produced with 500 replicates generated by resampling from the data. The boxplots of the modified Anderson Darling statistic AD2R suggest that the GEV-PWM right tail fit is the best one, followed by the GEV-MLE and the TCEV right tail fits.

5. Conclusion

Our simulation results revealed that the GEV distribution is more flexible in fitting data with a heavy right tail than the other generalizations of the Gumbel distribution. The TCEV distribution can also be a good choice. An application to an extreme wind speed data set in Florida confirmed the simulation study, with the GEV and TCEV models providing better fits than those of the other distributions.

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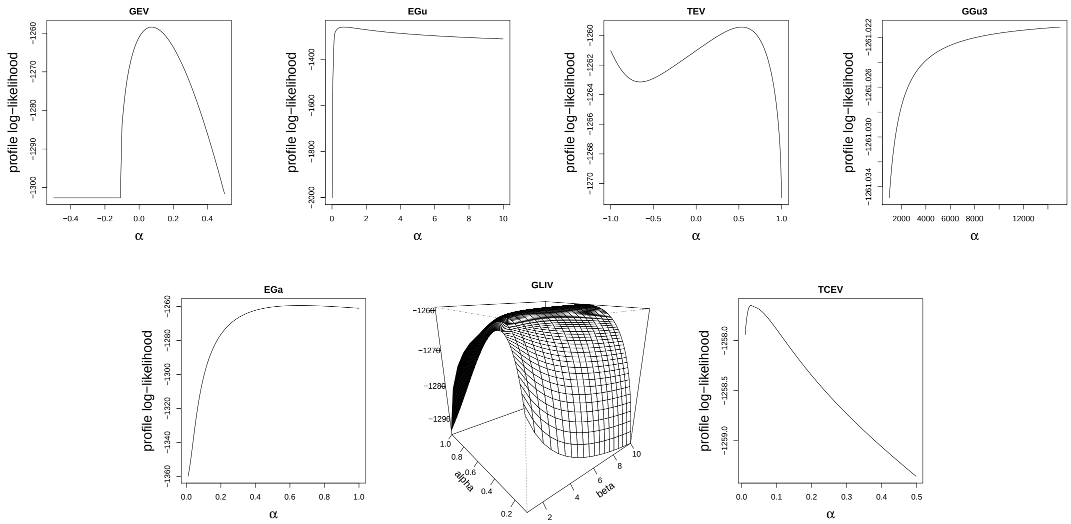

Figure 3: Profile log-likelihood - Monthly maximum wind speed data.

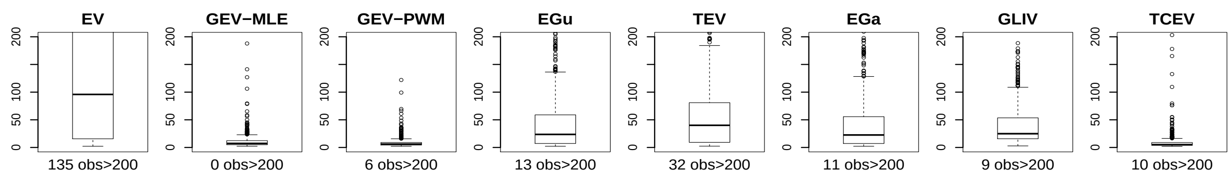

Figure 4: Boxplots of AD2R - random samples generated by resampling from the wind speed data.

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